Movie-Level Joint Inference for Open-Ended Long-Video QA

Abstract. Open-ended long-video QA requires answers to remain consistent across a film's characters, relationships, events, and ending state. We treat the movie as the inference unit: a single request combines timestamped subtitles, chronological visual observations, and the complete question set to generate one answer bank. The final SLoMO submission ranked first with 66.36% accuracy and a 3.02 score.

Keywords: long-video QA · movie-level inference · open-ended QA

1 System Overview

MovieQA and TVQA established story-centered video QA with movie narratives, dialogue, and visual evidence [3, 8]. LongVideoBench and EgoSchema extend evaluation to long-horizon video-language understanding [5, 9]. Video-ChatGPT and Video-LLaMA demonstrate instruction-tuned video-language models [4,10]. MovieChat and TimeChat introduce long-video memory and time-sensitive representations [6, 7]. Qwen-VL provides a versatile vision-language baseline [1]. The SF20K Competition Summary identifies information selection and narrative reasoning as central determinants of story-level QA [2]. Our system treats each film as one context unit and produces one coherent answer bank for its complete question set. Timestamped ASR, chronological visual observations, and all same-movie questions are assembled into one joint request; the answerer returns a question-ID keyed JSON bank for conversion to the submission CSV. Each request carries the movie ID, evidence packet, and complete question set. The returned bank preserves the original IDs and forms the complete submission unit for that movie.

Component	Final configuration
Answer model	GPT-5.6-Luna (`gpt-5.6-luna`)
API	OpenAI-compatible Chat Completions endpoint
Training	Zero-shot inference, no fine-tuning
Transcript	faster-whisper large-v3-turbo VTT with cue start times
Visual evidence	16 centered uniform chronological frames, 512 px, JPEG quality 82
Generation	Temperature 0, high reasoning effort, 5,000 output tokens, JSON bank
Runtime	Python $\geq$3.10, OpenAI SDK, HTTPX, OpenCV, and Pillow

2 Inference Protocol

For movie m, shared evidence E_m and the complete question set Q_m map to one bank $\{a_{m,1}, \ldots, a_{m,n_m}\}$ in original question order. Audio is transcribed with faster-whisper large-v3-turbo into timestamped VTT cues. We retain cue start times and the first and last 40,000 characters when needed. For a video with N frames, the i-th observation is $\min(N-1, \lfloor (i+0.5)N/16 \rfloor)$ for $i \in \{0, \ldots, 15\}$, then resized to a 512-pixel maximum side and JPEG encoded at quality 82. GPT-5.6-Luna receives the transcript, all ordered frames, and all movie questions in one zero-shot request. It silently builds a fact ledger and timeline before returning concise JSON answers with temperature 0, high reasoning effort, and a 5,000-token budget.
The textual packet places the timestamped transcript before the question list. The 16 JPEG images follow in chronological order, each with its timestamp. The response parser normalizes one concise answer for every requested question ID, so a complete answer bank is available directly to the submission writer.

3 Results

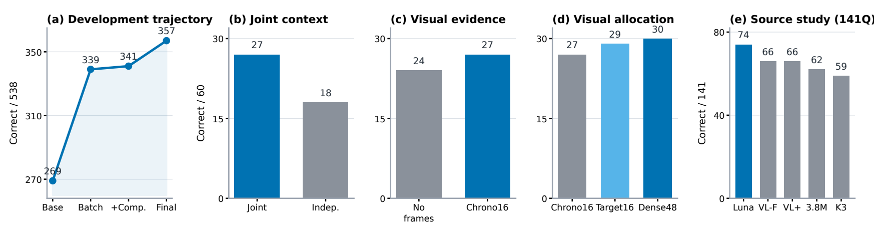

Figure 1. Evidence dashboard. (a) Development score trajectory. (b) Joint context. (c–d) Visual configurations. (e) Source comparison on the common 141-question overlap set. Values are correct answers.

The final system answers 357 of 538 questions and ranks first at 66.36% accuracy with a 3.02 score. Figure 1(a) tracks the progression from 269 to 339, 341, and 357 correct answers as the movie-level answer bank becomes the final prediction source.
Joint movie answering reaches 27/60, compared with 18/60 for independent calls. The no-frame condition reaches 24/60, chronological frames reach 27/60, and Target16 and Dense48 reach 29/60 and 30/60. These conditions isolate the contribution of shared movie context and visual evidence allocation under the same answerer.
On the common 141-question overlap set, GPT-5.6-Luna reaches 74 correct answers. Qwen3-VL Flash and Qwen-VL Plus each reach 66, Qwen3.8-Max reaches 62, and Kimi-K3 reaches 59. All answers for a film are generated in one response, retaining entity names, relationship states, event order, and ending state in one shared context before mapping the bank to official submission rows.

References

1. Bai, J., Bai, S., Yang, S., Wang, S., Tan, S., Wang, P., Lin, J., Zhou, C., Zhou, J.: Qwen-VL: A versatile vision-language model for understanding, localization, text reading, and beyond. arXiv preprint arXiv:2308.12966 (2023), https://arxiv.org/abs/2308.12966
2. Ghermi, R., Wang, X., Kalogeiton, V., Laptev, I.: SF20K competition 2025: Summary and findings. arXiv preprint arXiv:2605.01496 (2026), https://arxiv.org/abs/2605.01496
3. Lei, J., Yu, L., Bansal, M., Berg, T.L.: TVQA: Localized, compositional video question answering. In: Proceedings of the Conference on Empirical Methods in Natural Language Processing (2018), https://arxiv.org/abs/1809.01696
4. Maaz, M., Rasheed, H., Khan, S., Khan, F.S.: Video-ChatGPT: Towards detailed video understanding via large vision and language models. In: Proceedings of the Annual Meeting of the Association for Computational Linguistics (2024), https://arxiv.org/abs/2306.05424
5. Mangalam, K., Akshulakov, R., Malik, J.: EgoSchema: A diagnostic benchmark for very long-form video language understanding. arXiv preprint arXiv:2308.09126 (2023), https://arxiv.org/abs/2308.09126
6. Ren, S., Yao, L., Li, S., Sun, X., Hou, L.: TimeChat: A time-sensitive multimodal large language model for long video understanding. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (2024), https://arxiv.org/abs/2312.02051
7. Song, E., Chai, W., Wang, G., Zhang, Y., Zhou, H., Wu, F., Chi, H., Guo, X., Ye, T., Zhang, Y., Lu, Y., Hwang, J.N., Wang, G.: MovieChat: From dense token to sparse memory for long video understanding. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (2024), https://arxiv.org/abs/2307.16449
8. Tapaswi, M., Zhu, Y., Stiefelhagen, R., Torralba, A., Urtasun, R., Fidler, S.: MovieQA: Understanding stories in movies through question-answering. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition. pp. 4631–4640 (2016). https://doi.org/10.1109/CVPR.2016.501
9. Wu, H., Li, D., Chen, B., Li, J.: LongVideoBench: A benchmark for long-context interleaved video-language understanding. In: Advances in Neural Information Processing Systems (2024), https://arxiv.org/abs/2407.15754
10. Zhang, H., Li, X., Bing, L.: Video-LLaMA: An instruction-tuned audio-visual language model for video understanding. In: Proceedings of the Conference on Empirical Methods in Natural Language Processing: System Demonstrations (2023), https://arxiv.org/abs/2306.02858